

Techno on the Brain: Rhythmic Scaffolding and Visual Temporal Order Memory

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Abstract

Dynamic Attending Theory, or DAT, proposes that human attention oscillates in time, where it phase-locks to rhythm in the environment, and processes information more efficiently at the peaks of that oscillation as opposed to troughs. When applied to memory, DAT predicts that synchronizing visual encoding with an auditory rhythmic beat should improve memory for events aligned with the onset of the beat. Recent behavioral tests of this prediction have returned increasingly small effects, with the most recent literature showing strong results in favor of a null, and the prediction has not been tested directly on memory for the temporal order of events. Fifty-nine UCSB undergraduates ($N = 59$) completed a two-alternative temporal order judgment task across four auditory conditions: Sync, Out-of-sync, Metronome, and Ambient. Results showed that auditory condition had no effect on accuracy, and neither did attentional control nor musical sophistication moderate the result. The predicted behavioral signature of Dynamic Attending Theory in temporal order memory did not appear, which follows other research in a growing pattern of null findings. Together, these results indicate that the perceptual claim of dynamic attending can hold while its predicted consequence for memory remains unsupported.

Introduction

Episodic memory allows for the retrieval of specific past events alongside their contextual details, with one foundational feature of those events is their order in time. Memory for this temporal dimension is supported by neural mechanisms that partially dissociate from those underlying recognition of the events themselves (Jiang et al., 2025). Additionally, auditory experience carries rhythmic structure with speech possessing prosodic regularities and music having metrical structure. Previous literature also proposes that rhythm shapes attention at encoding, with some consequences for memory (Jones, 1976; Large & Jones, 1999). Our study asks whether this rhythmic scaffolding extends to memory for the order of visual events.

Dynamic Attending Theory, proposed by Jones (1976), provides a framework in which this question can be made precise, with the theory resting on three claims that build onto one another. The first concerns attention: perceptual awareness rises and falls in cycles, producing predictable peaks and troughs over time. The second concerns entrainment: when an isochronous beat is presented in the environment, internal attentional oscillators phase-lock to it, such that peaks of attentional readiness become temporally aligned with beat onsets. The third concerns processing consequences: stimuli arriving at attentional peaks are processed more efficiently than stimuli arriving in troughs, with measurable benefits to perception and downstream encoding. Together, these claims generate a specific prediction about how rhythmic structure should shape encoding. The memory prediction follows by combining DAT's three claims with one additional fact: memory depends on attention at encoding. If attentional peaks align with the onset of an external beat, and if stimuli arriving at peaks are processed more efficiently than stimuli arriving in troughs, then memory should be stronger for rhythmically aligned stimuli than for misaligned ones. This is the behavioral signature DAT predicts in episodic memory.

Reviewing the literature, there are behavioral experiments of DAT's memory prediction that follow a discernible pattern, with earlier work reporting positive effects of modest size. Firstly, Thavabalasingam et al. (2016) had participants encode sequences of visual scenes in which the timing between items was either rhythmically regular or randomized. Scenes encoded under structured/regular timing were recognized more accurately, with a Cohen's d ranging from 0.33 to 0.55 across three experiments. Additionally, Johndro et al. (2019) had participants encode standardized, emotionally neutral faces while listening to a lyricless instrumental track with a strict 4/4 structure. Faces presented in synchrony with the beat were better recognized than faces presented out of synchrony (Cohen's d of approximately 0.30 to 0.34), and both of these findings supported the DAT prediction at modest effect sizes. The picture changed under tighter controls, however. Hickey et al. (2020b) applied a similar on-beat versus off-beat manipulation with a 75 BPM track and obtained a full-sample null on recognition memory. A positive effect did emerge in a subgroup of participants who showed a behavioral pattern, but it was an analysis the authors flagged as exploratory. A succeeding registered report by Jones et al. (2023) tested the prediction with a pre-specified analysis, with 54 participants encoding sequences of visual scenes whose timing was either rhythmically predictable or arrhythmically variable, then tested recognition memory for said scenes. The resulting Bayes factor was 0.11, indicating substantial evidence in favor of the null hypothesis. ERP analyses also revealed distinct neural activity at encoding, but these neural differences did not produce differential recognition memory. The dissociation between neural sensitivity to temporal structure and behavioral memory outcomes raised questions about whether item recognition is the right behavioral measure for detecting DAT's predicted effects.

Across the four studies above, a declining pattern is visible, with reported effects having shrunk as designs have tightened and the most recent result being a registered report producing clear Bayesian evidence in favor of the null. A reasonable reading of this trajectory is that DAT's predicted behavioral signature in episodic memory may not exist, or exists only at a magnitude that current studies cannot detect. One feature of the existing literature complicates this claim. Each of the behavioral tests described above used item recognition as the dependent variable, where item recognition asks whether a particular stimulus appeared in the original sequence. Episodic memory has another component: temporal order memory, which concerns the relative timing of stimuli rather than the confirmation of their presence. This dimension has theoretical relevance for DAT, since the neural mechanism most often invoked for rhythmic processing, cross-frequency phase-amplitude coupling, has been linked specifically to temporal order memory (Jiang et al., 2025; Townsend et al., 2024; Wang et al., 2024). This mechanism links DAT to the dimension in which a rhythmic encoding benefit is most directly predicted. These nulls therefore depend on whether item recognition is the correct measure to detect DAT's predicted effect. If it is not, the downward trajectory in recent literature may reflect a mismatch between the theoretical prediction and the dependent variable used to test it. Testing the prediction on temporal order memory is a direct way to determine which of these readings is correct or not.

The present study addresses this gap within a single within-subjects design. Participants completed a two-alternative forced choice temporal order judgment task. After encoding a sequence of images, they were shown two images from that sequence and asked to judge which appeared earlier. This task isolates the memory dimension that has not been directly tested in the behavioral literature on DAT. The design's primary contrast, between the Sync and Out-of-sync

conditions, uses a single audio file shifted in time, so that loudness, tempo, spectral content, beat regularity, and predictability are held identical across the two conditions. The only difference between them is the phase relationship between the audio and the visual stimuli: in Sync, image onsets coincide with the beat; in Out-of-sync, image onsets fall between beats. Recognizing that music engages cognitive and emotional processes that extend beyond rhythmic structure, two additional conditions were included to separate these confounds. The Metronome condition presents isochronous clicks at the same tempo as the sync and out of sync condition, isolating rhythmic structure from musical content. The ambient condition presents a continuous non-rhythmic ambient drone, isolating auditory presence from rhythmic structure. If a Sync versus Out-of-sync effect emerges, comparison against these conditions clarifies whether the effect derives from rhythmic alignment or from broader features of the auditory environment.

The primary prediction follows from DAT's phase-alignment claim. We predict that accuracy in the Sync condition will be higher than the accuracy in the Out-of-sync condition. In Sync trials, image onsets coincide with the beat, the moment at which DAT predicts attentional readiness to peak; in Out-of-sync trials, image onsets fall between beats. The remaining condition contrasts are interpretive rather than predictive: Sync versus Metronome assesses whether musical content adds anything beyond bare rhythmic structure, and Sync versus Ambient assesses whether rhythmic structure adds anything beyond mere auditory presence. The lag manipulation embedded within the task generates a secondary prediction that serves as a manipulation check on task sensitivity, where accuracy is predicted to increase as the distance between target and foil in the encoding sequence increases, since items separated by greater distances are more discriminable in memory.

Two pre-specified individual difference measures are predicted to moderate the condition effect. The Attentional Control Scale (ACS) is included to test whether individual differences in attentional control predict rhythmic memory effects. Hickey et al. (2020b) found that the strength of participants' neural entrainment to the beat predicted the magnitude of the rhythmic memory benefit, and a follow up analysis of the same dataset linked the benefit to post-perceptual attentional ERP components (Hickey et al., 2020a). The ACS provides a behavioral test of whether trait-level attentional control plays a similar role. The Goldsmiths Musical Sophistication Index (Gold-MSI) is included to revisit a moderation hypothesis that Johndro et al. (2019) tested directly and failed to support on item recognition. Whether musical training moderates rhythmic memory effects on temporal order remains untested. These moderator analyses ask whether DAT's predicted behavioral signature depends on stable individual differences in attention or musical training.

Design

The experiment used a within-subjects design with auditory condition (Sync, Out-of-sync, Metronome, Ambient) as a four-level factor. Each participant completed 40 trials of a two-alternative temporal order judgment task, with 10 trials per condition. Accuracy on the temporal order judgment served as the dependent variable. Within each trial, target-foil lag (the distance in the encoded sequence between the two test items) varied. For analysis, trials were categorized into three lag levels (Short, Medium, Long), creating a second within-subjects factor. Trial order was randomized for each participant.

Participants

Fifty-nine UCSB undergraduate students participated in the study ($N = 59$). Participants were recruited through the UCSB's SONA participant pool and tested in person between March and April 2026. Compensation consisted of 1 SONA course credit or \$10. All 59 participants completed the experimental task and were included in the primary analyses. No participants were excluded on the basis of response time or accuracy.

Apparatus

All participants were tested in person in the Psychology East basement room 0814C (within the Miller Lab). The experiment was presented using jsPsych, where participants viewed the visual stimuli on a 60Hz desktop monitor and listened to auditory stimuli through over-ear headphones. Responses were collected using the F and J keys to determine answer choices for the task.

Stimuli

Visual stimuli were 440 photographs drawn from the MIT Indoor Scenes Dataset (Quattoni & Torralba, 2009), a publicly available image set of indoor environments spanning multiple scene categories. Of these, 400 images served as encoding stimuli for the main trials and 40 served as encoding stimuli for the practice trials. Each image was presented for 933.34 ms (~933 ms), corresponding to exactly two beats of the beat structure described below. The Sync and Out-of-sync conditions used the same underlying 4-on-the-floor techno track, composed for the experiment in Ableton Live using a Roland TR-909 drum kit (kick, ride cymbal, and bass with minimal equalization). The track had an inter-beat interval of 466.67 ms (~467 ms; tempo 128.57 BPM, ~129 BPM). The visual stimuli appeared at exactly half the beat rate, meaning one image appeared for every two beats. In the Sync condition, image onsets coincided with beat onsets. In the Out-of-sync condition, the audio was shifted by 233 ms (half an inter-beat interval), placing image onsets at the half-beat midpoint between kicks. This shift held tempo, spectral content, predictability, and overall loudness constant across the two conditions, with the only systematic difference being the phase relationship between audio and visual onsets. The Metronome condition presented bare isochronous clicks at the same tempo. The ambient condition presented a continuous non-rhythmic ambient track lacking discernible beat structure. Audio was triggered at the onset of the first image of each trial and ceased at the end of the 10th image, such that participants heard approximately 9.33 seconds (~10 seconds) of audio per trial.

Measures

Attentional Control Scale

The Attentional Control Scale (Derryberry & Reed, 2002) is a 20-item self-report measure of trait attentional control, assessing individual differences in the capacity to focus, shift, and divide attention. Each item is rated on a 4-point Likert scale (1 = "almost never" to 4 = "always"). Items were reverse-scored as specified by Derryberry and Reed (2002), then summed to yield a total score ranging from 20 to 80, with higher scores indicating greater attentional control.

Goldsmiths Musical Sophistication Index

The Goldsmiths Musical Sophistication Index (Müllensiefen et al., 2014) is a self-report measure of musical sophistication, capturing individual differences in active musical engagement, perceptual abilities, and musical training. The version administered here included 31 self-report items rated 7 musical-training items with item-specific response scales (e.g., years of musical training, hours of weekly practice, etc). Higher total scores indicate greater musical sophistication.

State-Trait Anxiety Inventory

The STAI-6 (Marteau & Bekker, 1992) is a 6-item short form of the State-Trait Anxiety Inventory designed to assess state anxiety, with each item is rated on a 4-point Likert scale from "not at all" to "very much so." Higher scores indicate greater state anxiety, and the STAI-6 was administered to characterize sample state anxiety.

Procedure

The on-screen instructions described the structure of each trial: a fixation cross appeared in the center of the screen, followed by a sequence of 10 images presented while audio played, then by two test images shown side by side. Participants were instructed to "pay attention to the order of the images" at encoding, and at test to press F or J to indicate which test image had appeared first, responding as quickly and accurately as possible.

Participants first completed two practice trials (one Sync, one Ambient) with trial-by-trial accuracy feedback. The main task consisted of 40 trials, and the condition order was randomized for each participant. On each trial, a fixation cross was first presented, followed by 10 images presented sequentially at the timing described in the Stimuli section (one image every 933.34 ms, two beats per image, approximately 9.33 seconds of encoding total). The corresponding audio for that trial's condition played during encoding and ceased at the end of the encoding phase. After the encoding sequence, two images from the encoded sequence appeared side by side, and participants made their temporal order judgment via the F or J key. A 500 ms blank screen preceded the next trial. Test images were drawn from positions 2 through 9 of the encoded sequence to avoid primacy and recency effects, with the target preceding the foil by at least two positions. Target-foil lag thus ranged from 2 to 7 and was categorized as Short ($\text{lag} \leq 2$), Medium ($2 < \text{lag} \leq 4$), or Long ($\text{lag} > 4$) for analysis.

Self-paced breaks were inserted after every 10 trials (i.e., after trials 10, 20, and 30). During breaks, participants saw a screen indicating their progress and could resume by pressing Continue. Following the 40 main trials, participants completed the three self-report questionnaires (the Attentional Control Scale, the Goldsmiths Musical Sophistication Index, and

the State-Trait Anxiety Inventory short form). On completion, participants viewed a debrief screen explaining the study purpose and received 1 SONA course credit or \$10 in compensation.

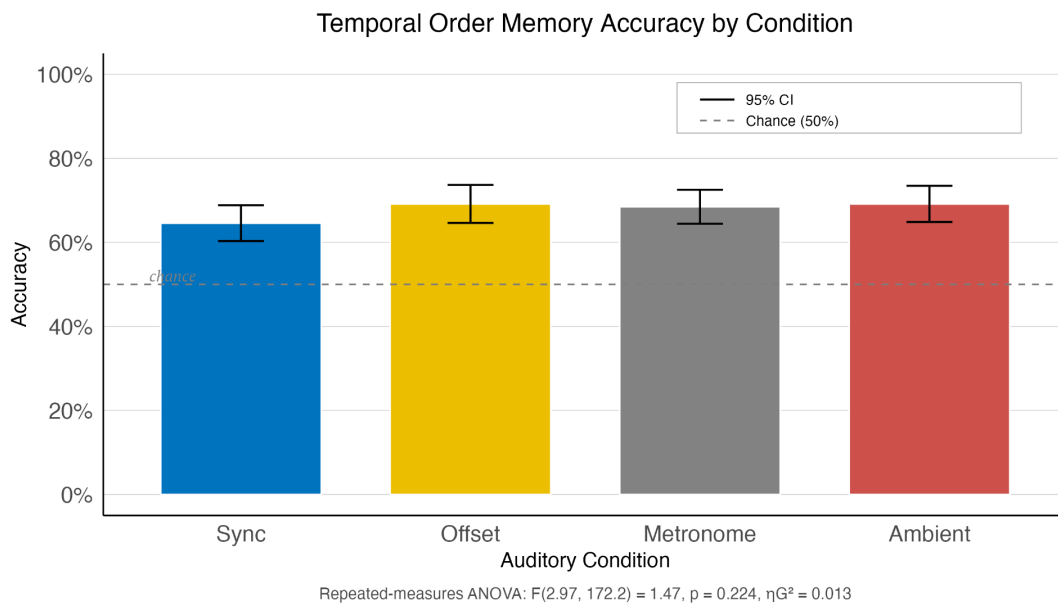
Data Analysis

All analyses were conducted in R and no trials were removed on the basis of response time or accuracy. The primary analysis was a one-way repeated-measures ANOVA on accuracy with condition (Sync, Out-of-sync, Metronome, Ambient) as the within-subjects factor. Pairwise comparisons among condition levels used Bonferroni correction. Two secondary repeated-measures ANOVAs examined the lag manipulation: a one-way ANOVA on accuracy across lag categories (Short, Medium, Long), serving as a manipulation check on task sensitivity, and a two-way ANOVA testing the Condition $\times$ Lag interaction. Generalized eta-squared was reported as the effect size. Moderator analyses tested whether condition effects varied with individual differences in attentional control (ACS) or musical sophistication (Gold-MSI). Each moderator was entered as a continuous predictor in a separate linear mixed-effects model, with condition, the mean-centered moderator, and their interaction as fixed effects and a random intercept for participants. Response time was logged for quality control only, and the STAI-6 was descriptive.

Results

Across all participants and conditions, mean accuracy on the temporal order judgment task was 67.84% ($SD = 11.18\%$, range 45.00% to 90.00%). Performance was significantly above the 50% chance level, $t(58) = 12.25$, $p < .001$, $d = 1.59$, indicating that participants successfully completed the temporal order judgment task at well above chance.

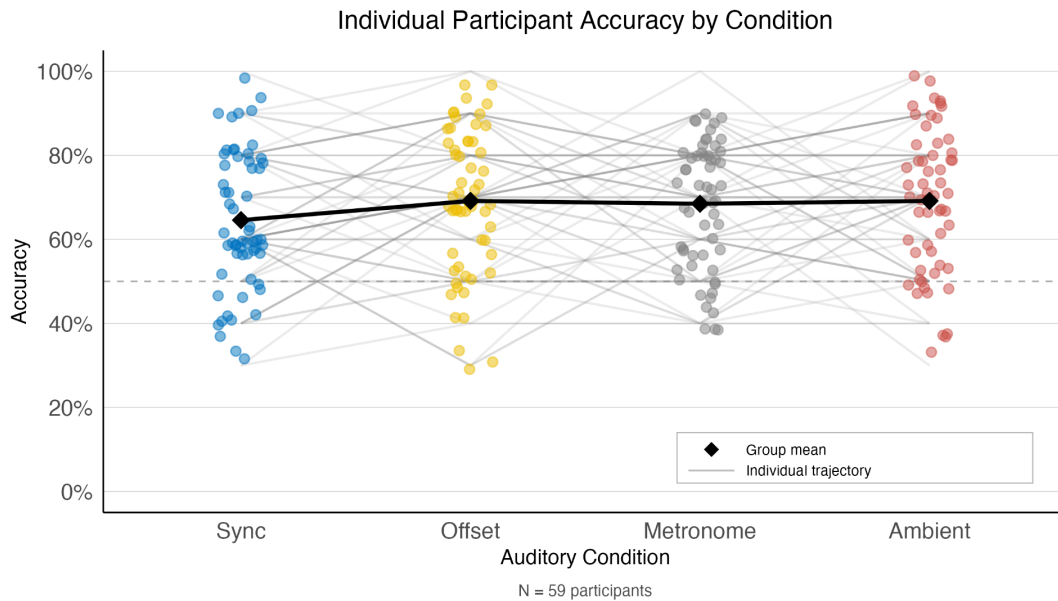
Mean accuracy was 64.58% in the Sync condition ($SD = 16.33\%$, 95% CI [60.32, 68.83]), 69.15% in the Out-of-sync condition ($SD = 17.35\%$, 95% CI [64.63, 73.67]), 68.47% in the Metronome condition ($SD = 15.52\%$, 95% CI [64.43, 72.52]), and 69.15% in the Ambient condition ($SD = 16.54\%$, 95% CI [64.84, 73.46]).



Primary Analysis: Effect of Condition

A one-way repeated-measures ANOVA on accuracy with condition (Sync, Out-of-sync, Metronome, Ambient) as the within-subjects factor revealed no significant effect of condition,

$F(2.97, 172.23) = 1.47, p = .224, \eta G^2 = .014$. Sync accuracy was numerically the lowest of the four conditions with Out-of-sync being the highest, opposite to the predicted direction.



Pairwise Comparisons

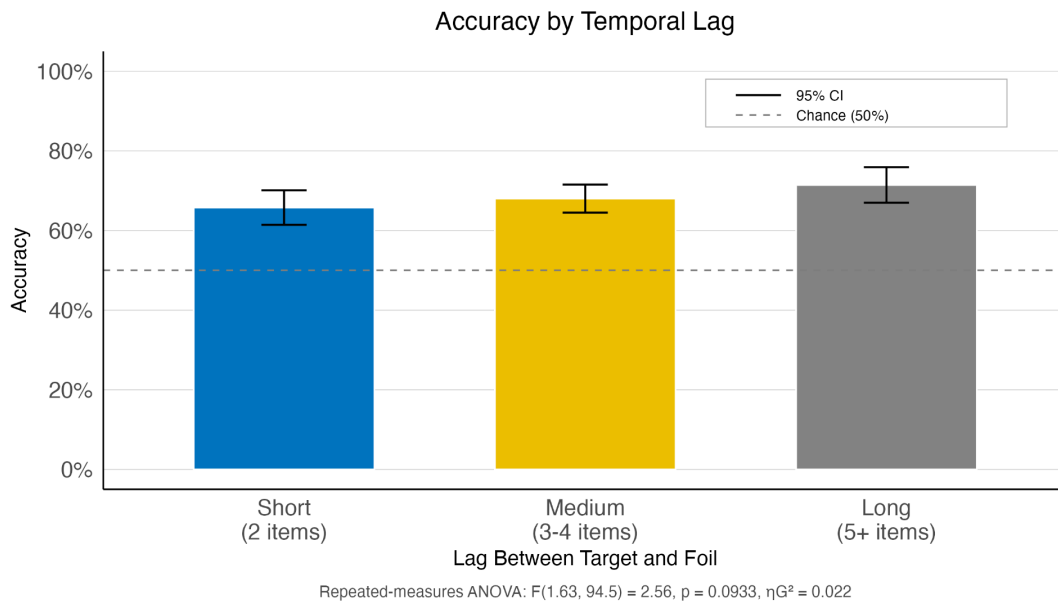
Bonferroni-corrected pairwise comparisons assessed each of the six condition pairs. The primary contrast between Sync and Out-of-sync was not significant: Sync accuracy was 4.58 percentage points lower than Out-of-sync accuracy, $t = -1.87, p = .398, d = -0.27$. Comparisons between Sync and the two control conditions were also not significant: Sync versus Metronome (mean difference = -3.90%, $t = -1.57, p = .737, d = -0.24$) and Sync versus Ambient (mean difference = -4.58%, $t = -1.79, p = .468, d = -0.28$).

Lag Manipulation

A one-way repeated-measures ANOVA on accuracy with lag category (Short, Medium, Long) as the within-subjects factor did not yield a significant effect of lag, $F(1.63, 94.50) = 2.56, p = .093$,

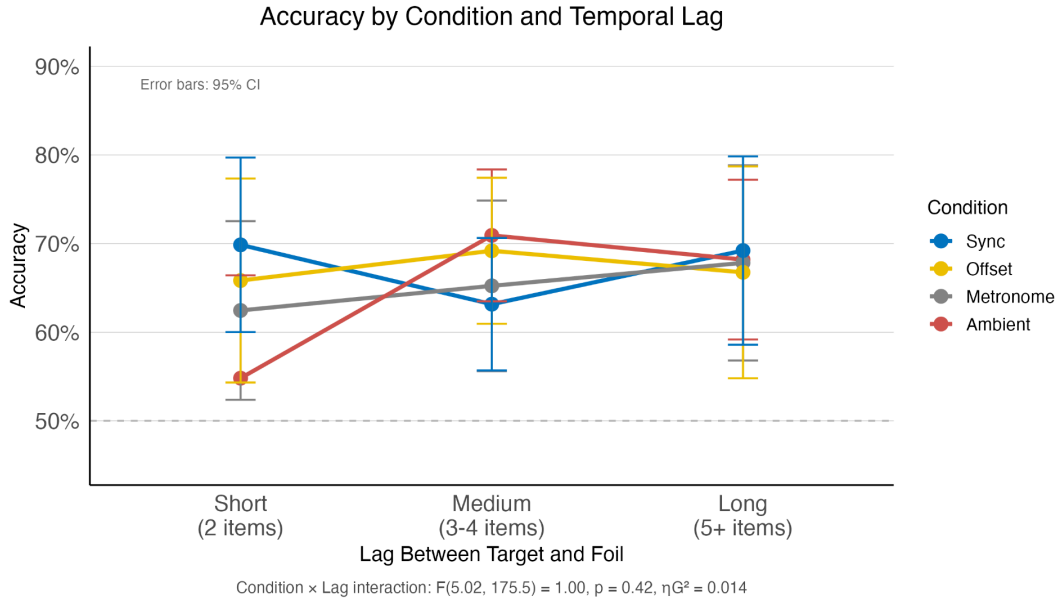
$\eta^2 = .022$. Mean accuracy increased monotonically with lag: 65.75% for Short trials ($SD = 16.64\%$), 68.01% for Medium trials ($SD = 13.54\%$), and 71.43% for Long trials ($SD = 17.10\%$).

The directional pattern was consistent with the predicted increase in accuracy with greater target-foil distance.



Condition $\times$ Lag Interaction

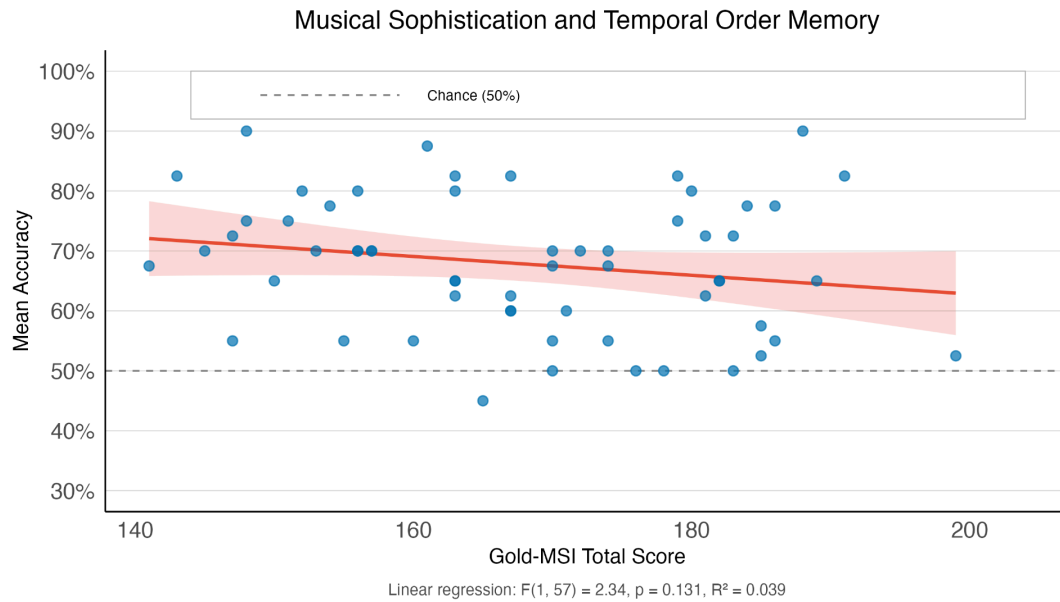
A two-way repeated-measures ANOVA with condition and lag category as within-subjects factors revealed no significant Condition $\times$ Lag interaction, $F(5.02, 175.54) = 1.00$, $p = .420$, $\eta^2 = .014$.



Moderator Analyses

Two pre-specified moderator analyses examined whether condition effects varied with individual differences in attentional control or musical sophistication. The Attentional Control Scale (ACS) total scores in the sample ranged from 39 to 61 ($M = 49.44, SD = 5.58, N = 57$). A linear mixed-effects model with condition, ACS (mean-centered), and their interaction predicting trial-level accuracy revealed no significant Condition $\times$ ACS interaction, $F(3, 165) = 1.41, p = .243$. Neither the main effect of condition ($F(3, 165) = 1.52, p = .211$) nor the main effect of ACS ($F(1, 55) < 0.01, p = .939$) reached significance.

The Goldsmiths Musical Sophistication Index (Gold-MSI) total scores ranged from 141 to 199 ($M = 167.92, SD = 14.08, N = 59$). A parallel linear mixed-effects model revealed no significant Condition $\times$ Gold-MSI interaction, $F(3, 171) = 1.25, p = .295$. The main effect of Gold-MSI was also non-significant, $F(1, 57) = 2.34, p = .131$, as was the main effect of condition, $F(3, 171) = 1.48, p = .222$.



State Anxiety

State anxiety, as measured by the STAI-6, was reported as descriptive context for the sample ($M = 6.10$, $SD = 3.91$, range 0-14, $N = 58$).

Discussion

This study tested whether rhythmic auditory stimulation synchronized with the onset of visual stimuli during encoding would improve memory for the temporal order of images. The results did not support this prediction: the primary analysis revealed that there was no effect of auditory condition on temporal order judgment accuracy, and the main contrast between the Sync and Out-of-sync conditions was nonsignificant. Neither the Metronome nor the Ambient comparison approached significance, and neither pre-specified moderator, attentional control (ACS) nor musical sophistication (Gold-MSI), interacted with condition. Accuracy in the Sync condition was numerically the lowest out of the four conditions, with the Out-of-sync condition surprisingly being the highest. Even though this difference was small, it did not reach significance and will not be interpreted further.

Several features of the data indicate that this null result depicts the absence of an effect rather than a failure to measure one. Participants averaged a mean accuracy of 67.84%, which is well above the 50% chance level, confirming that the temporal order judgment task captured genuine memory for sequence position and participants weren't relying on chance. Accuracy also increased in relation to a larger lag manipulation, meaning participants displayed higher accuracy when the distance between images were larger in comparison to shorter, which is consistent with our prediction that items separated by greater distances in the encoded sequence are easier to discriminate. The task therefore measured what it was designed to measure, and participants were engaged throughout encoding and retrieval.

To situate this in the context of the previous literature, this result is best understood as the most recent point in an already decreasing empirical trajectory. Early behavioral tests of dynamic attendance in episodic memory reported positive effects of modest size with Thavabalasingam et al. (2016) and Johndro et al. (2019), both with effect sizes in the small range. As experimental controls were tightened, these effects diminished: Hickey et al. (2020b) obtained a full-sample null on recognition memory under a matched on-beat vs. off-beat manipulation, and Jones et al. (2023), in a registered report with a pre-specified Bayesian analysis plan, reported clear evidence for a strong null. The present study extends this trajectory by applying the cleanest acoustic control yet used in the literature by holding tempo, spectral content, predictability, and loudness constant by shifting a single audio file in time, testing temporal order memory. The predicted effect did not appear, meaning this study is the third null in a literature whose positive findings were small at the outset.

These results have implications for how dynamic attending theory should be applied to episodic memory. DAT is well supported as an account of perception and timing, where rhythmic context has been shown to modulate the detection and discrimination of stimuli that fall on or off the beat. What the present study tested was *an extension* of that claim: the prediction that attentional entrainment during encoding leaves a noticeable/measurable trace in the memory representation, such that information aligned with attentional peaks is retained more accurately. It is this extension, rather than dynamic attending itself, that our null results dispute. The extension requires a specific causal chain: attentional readiness must align with the beat, that alignment must change how stimuli at the peak are processed, and the difference in processing must produce a difference in what is later remembered. The first link of this chain has strong empirical support in perceptual tasks. The third link, where processing efficiency at encoding translates

into a memory improvement at the retrieval stage, is where the literature has accumulated nulls. The empirical foundation for this step is narrower than it first appears, since of the two original positive findings, Thavabalasingam et al.,’s result (2016) has failed to replicate under direct retest (Kulkarni & Hannula, 2021), and the Johndro et al.,’s (2019) results have not been tested under the matched acoustic controls. The entrainment of attention by a rhythmic context and the consequence of that entrainment on memory are separable claims, and the evidence for the first does not guarantee the second.

The conclusion that rhythmic synchrony does not improve temporal order memory should be distinguished from the more general claim that rhythm has no role in memory for sequences. Ren et al. (2024) found that the schematic structure and familiarity of a musical background influenced how accurately participants learned and retrieved visual sequences. However, that effect emerged under conditions that differ from the present study in two ways that matter: Memory in their task was built over repeated exposures, with each sequence encoded multiple times, so that performance reflected statistical learning rather than a single encoding episode. Also, the effective variable was the familiarity and regularity of the musical structure, not the temporal alignment of sounds with visual onsets. These are not competing predictions about the same mechanism. One concerns whether accumulated knowledge of a musical context shapes how sequences are extracted over time, while the other concerns whether a single encoding event is improved when its timing coincides with a techno beat. Rhythm-related context can shape sequence memory through the first mechanism; the design in this study isolated the second, and produced no effect.

Limitations

First, one main theory surrounding techno music's "hypnotic" and "trance-like" effect is the repetition of an isochronous beat for an extended period of time. Only playing around 10 seconds of the beat before cutting the audio might not allow enough time for the "hypnosis" effect to acclimate, and the techno beat may have acted simply as background noise rather than a driving feature shifting and influencing attention. Second, the study included no measure of whether participants noticed the timing relationship between the sounds and the images, and the possibility raised by Kulkarni and Hannula (2021), that explicit awareness may be required for an effect to emerge, cannot be evaluated with our data. Additionally, our manipulation used a single tempo (~129 BPM) and a single set of stimuli, so the results do not speak to whether a benefit might emerge at other rates, or with other materials/modalities. A sensitivity analysis confirmed that with $N = 59$, the design had 80% power to detect a condition effect of $\eta^2 \geq .043$ and a Sync versus Out-of-sync difference of $d_z \geq 0.37$. The observed effects were smaller than both thresholds, so smaller true effects cannot be ruled out. The positive effects reported in previous literature were themselves small, with standardized effect sizes nearing 0.30, which places any true effect in a range that would require a larger sample to detect reliably.

Future Direction

These limitations can be addressed through pursuing more research on techno music and its effects on memory and attention. Firstly, by extending the duration of the rhythmic audio beyond a single encoding sequence, or presenting it continuously across multiple trials, would test whether longer exposure is required for any rhythmic effect on memory to develop. For example, a longer task would be provided with minimal to no delays or opportunities for breaks to test the amount of time needed for techno music to have an effect on the brain. Including an aspect to test

awareness would let the role of explicit knowledge of the timing relationship be assessed rather than assumed. Additionally, manipulations across a range of tempos and stimulus types would establish whether the absence of a benefit is generalized or is specific to the techno beat tested in this study.

Across the well-controlled tests conducted in this literature, including ours, the predicted behavioral signature of dynamic attending in episodic memory has not appeared. Rhythmic context reliably shapes perception, and it shapes sequence memory when structure is learned over repeated exposure, but the proposal that synchronizing attention to a beat during a single encoding episode improves the resulting memory remains, at present, without empirical support.

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